Unauthenticated vs Authenticated Scan

In this lab, we compare the results of unauthenticated and authenticated scans taken in Win11 OS.

Setup

- **VM**: win11-25h2-pro
- **Scanner**: Tenable Nessus

Things we do

We are going to perform an unauthenticated and an authenticated scan on our Win11 virtual machine with an internal scan server, which are both located in the same virtual network. Compare the reports generated after the scannings.

Difference

- **Unauthenticated Scans**: Vulnerability Management tools do not possess your admin credential. Only perform surface-level evaluation. Mainly focus on external facing vulnerability findings, which is what usually attackers see.
- **Authenticated Scans**: Scanners perform in-depth vulnerability scans with a provisioned admin credential, going through the OS configuration, file system, etc.

Step

- **Add firewall rule**: Since we are doing this lab in Azure environment, Azure provides NSG in the virtual network, we need to add a firewall rule which ensure the traffic between the scanner and the target. Allow all the in-bound traffic coming from IP 10.0.0.8 which is our internal Tenable Nessus server.
- **Start Scans**: Unauthenticated scan took 10 mins, authenticated scan took 23 mins.

Final Report

Summary

Critical	High	Medium	Low	Info	Total
0	0	4	1	33	38

Details

Severity	Plugin Id	Name
Medium	157288	TLS Version 1.1 Deprecated Protocol
Medium	51192	SSL Certificate Cannot Be Trusted
Medium	57582	SSL Self-Signed Certificate
Medium	104743	TLS Version 1.0 Protocol Detection
Low	10114	ICMP Timestamp Request Remote Date Disclosure
Info	10785	Microsoft Windows SMB NativeLanManager Remote System Information Disclosure
Info	110723	Target Credential Status by Authentication Protocol - No Credentials Provided
Info	86420	Ethernet MAC Addresses
Info	277654	TLS Supported Groups
Info	106716	Microsoft Windows SMB2 and SMB3 Dialects Supported (remote check)
Info	45590	Common Platform Enumeration (CPE)
Info	138330	TLS Version 1.3 Protocol Detection

Summary

Critical	High	Medium	Low	Info	Total
0	5	4	2	136	147

Details

Severity	Plugin Id	Name
High	193266	Security Updates Outlook for Windows (April 2024)
High	298646	Microsoft Windows Notepad < 11.2510 Command Injection (February 2026)
High	250276	Microsoft Teams for Desktop < 25122.1415.3698.6812 Remote Code Execution (August 2025)
High	103569	Windows Defender Antimalware/Antivirus Signature Definition Check
High	166555	WinVerifyTrust Signature Validation CVE-2013-3900 Mitigation (EnableCertPaddingCheck)
Medium	104743	TLS Version 1.0 Protocol Detection
Medium	57582	SSL Self-Signed Certificate
Medium	157288	TLS Version 1.1 Deprecated Protocol
Medium	51192	SSL Certificate Cannot Be Trusted
Low	10114	ICMP Timestamp Request Remote Date Disclosure
Low	264898	Microsoft Teams for Desktop < 25163.3611.3774.6315 Elevation of Privilege (July 2025)